Clinical Risk Modeling and Explainability: Mortality Prediction in Heart Failure Patients

M2 Data Science Project Report · Tabular Clinical Data, Gradient Boosting and SHAP Attribution · October 2026

Abstract

This report describes the construction and evaluation of a supervised learning system that estimates the risk of death among patients with heart failure from routinely collected clinical measurements. Using the Heart Failure Clinical Records dataset (299 patients, 11 predictors after exclusions), a regularised logistic regression baseline was compared with a LightGBM gradient-boosted tree classifier under a stratified 80/20 hold-out protocol. LightGBM reached a test ROC-AUC of 0.759 against 0.734 for the baseline, while the baseline obtained the higher PR-AUC (0.591 against 0.542). Platt scaling reduced the Brier score of the boosted model from 0.192 to 0.186, an improvement that remained short of the 0.15 target. SHAP attribution identified serum creatinine, ejection fraction and age as the dominant drivers of predicted risk, which is consistent with established clinical knowledge. The ROC-AUC (0.85) and Brier score (0.15) thresholds set in the project specification were not reached; the report discusses the small sample size as the principal explanation and lists the parts of the specification that remain to be completed.

1. Introduction

Early identification of patients at high risk of an adverse outcome allows clinical teams to concentrate monitoring and treatment where it is most needed. In this setting a predictive model is useful only if three conditions hold: it discriminates well between patients who will and will not experience the event, its predicted probabilities correspond to observed event frequencies, and its outputs can be explained to a clinician in terms of the patient's own measurements.

The project specification therefore defines four phases: data preparation, modeling with a baseline comparison, explainability through SHAP, and deployment as a REST service. It sets quantitative targets of ROC-AUC of at least 0.85, a Brier score of at most 0.15, calibration by Platt scaling, and a single-sample inference latency of at most 15 ms. Because false negatives are more costly than false positives in a clinical context, and because the outcome classes are imbalanced, accuracy is not used as a performance measure in this work.

The work reported here covers the first three phases and the serialisation of the model. The deployment phase and hyperparameter optimisation were not carried out (Section 6).

2. Data

2.1 Source and structure

The analysis uses the Heart Failure Clinical Records dataset distributed on Kaggle, which originates from the study of Ahmad et al. (2017) and was subsequently analysed by Chicco and Jurman (2020). It contains 299 patients and 13 columns: 11 clinical predictors, a follow-up period in days (`time`), and the binary outcome `DEATH_EVENT`. The dataset was selected from the two options recommended in the specification because it is clean, tabular and small enough to be processed on a laptop CPU in seconds.

The continuous variables are age, creatinine phosphokinase (CPK), ejection fraction, platelet count, serum creatinine and serum sodium. The binary variables are anaemia, diabetes, high blood pressure, sex and smoking status. No missing values were found in any column, so no imputation was required, and all categorical variables were already coded as 0/1, so no further encoding was necessary.

2.2 Outcome distribution

Of the 299 patients, 96 (32.1%) died during follow-up and 203 (67.9%) survived. The classes are therefore moderately imbalanced. A classifier that ignores the predictors would attain a PR-AUC of approximately 0.32 and a Brier score of approximately 0.218 (the product of the two class proportions), and these values serve as reference points for the results below.

2.3 Exploratory analysis

Group means by outcome (Table 1) indicate that patients who died were older, had lower ejection fraction and higher serum creatinine, and had somewhat lower serum sodium and platelet counts.

Variable	Survived (n = 203)	Died (n = 96)
Age (years)	58.76	65.22
Creatinine phosphokinase (mcg/L)	540.05	670.20
Ejection fraction (%)	40.27	33.47
Platelets (kiloplatelets/mL)	266,657	256,381

Variable	Survived (n = 203)	Died (n = 96)
Serum creatinine (mg/dL)	1.18	1.84
Serum sodium (mEq/L)	137.22	135.38

Table 1. Mean value of the continuous predictors by outcome.

The linear correlations with `DEATH_EVENT` are modest. The largest in absolute value are follow-up time (−0.53), serum creatinine (+0.29), ejection fraction (−0.27), age (+0.25) and serum sodium (−0.20). Diabetes, sex and smoking are essentially uncorrelated with the outcome ($|r| \leqslant 0.01$). Several continuous variables are strongly right-skewed: CPK has a median of 250 and a maximum of 7,861, and serum creatinine has a median of 1.1 and a maximum of 9.4. This skewness is of little consequence for tree-based models, which are invariant to monotone transformations, but it is relevant to the logistic regression baseline.

2.4 Exclusion of the follow-up variable

The variable `time` has the strongest association with the outcome but was removed from the feature set. It records the length of follow-up, which is shortened by death itself, and it would not be known at the moment a risk estimate is required. Retaining it would introduce target leakage and would produce performance figures that could not be reproduced in practice. All models below therefore use the 11 remaining predictors.

3. Methodology

3.1 Data partitioning

The data were divided into training (239 patients) and test (60 patients) sets using a stratified 80/20 split with a fixed random seed (42), so that the outcome proportion is preserved in both subsets. The test set was not used for any model-fitting decision.

3.2 Models

Two classifiers were compared.

Baseline. A logistic regression with balanced class weights, preceded by standardisation of the predictors (zero mean, unit variance) in a single scikit-learn pipeline. Scaling is necessary for this model because the predictors differ by several orders of magnitude (platelets are of the order of 10^5, serum creatinine of the order of 1).

Gradient-boosted trees. A LightGBM classifier (Ke et al., 2017) with 100 trees, maximum depth 3, learning rate 0.05 and a positive-class weight equal to the ratio of negative to positive training cases (`scale_pos_weight`). The shallow depth and low learning rate were chosen a priori as a conservative setting for a small sample; no search over hyperparameters was performed.

3.3 Probability calibration

Class weighting improves sensitivity to the minority class but shifts predicted probabilities away from empirical frequencies. The LightGBM model was therefore wrapped in `CalibratedClassifierCV` with sigmoid (Platt) scaling (Platt, 1999) and three internal folds on the training set, in accordance with the specification.

3.4 Evaluation metrics

Discrimination was measured by the area under the ROC curve (ROC-AUC) and by the area under the precision-recall curve, estimated as average precision (PR-AUC). Probability quality was measured by the Brier score, the mean squared difference between predicted probability and observed outcome, where lower values are better. All metrics were computed on the held-out test set.

3.5 Explainability

Feature attributions were computed with SHAP (Lundberg and Lee, 2017) using the exact `TreeExplainer` on the fitted, uncalibrated LightGBM model, evaluated on the test set. Global behaviour is summarised by a beeswarm plot, and a local explanation is given by a waterfall plot for the test patient with the highest predicted risk. SHAP values for this model are expressed in log-odds units.

4. Results

4.1 Model comparison

Model	ROC-AUC	PR-AUC
Logistic regression (baseline)	0.734	0.591
LightGBM	0.759	0.542
LightGBM, calibrated (Platt)	0.741	0.511

Table 2. Discrimination on the test set (n = 60).

LightGBM obtained a higher ROC-AUC than the baseline (0.759 against 0.734), whereas the baseline obtained a higher PR-AUC (0.591 against 0.542). The two models are therefore not clearly separable, and with 60 test observations neither difference should be regarded as established. All three models exceed the no-skill PR-AUC of about 0.32, so the predictors carry real signal. Calibration lowered both ROC-AUC (0.759 to 0.741) and PR-AUC (0.542 to 0.511); this is plausible because the calibrated model is fitted on resampled subsets of an already small training set, but the present experiment does not isolate the cause.

4.2 Calibration

Model	Brier score
LightGBM, uncalibrated	0.1919
LightGBM, Platt-calibrated	0.1862
No-skill reference (prevalence)	≈ 0.218

Table 3. Brier score on the test set (lower is better).

Platt scaling reduced the Brier score by 0.006. The calibrated value is below the no-skill reference, which confirms that the probabilities are informative, but it remains above the specification target of 0.15.

4.3 Global explanation

The SHAP beeswarm plot ranks the predictors by mean absolute contribution as follows: serum creatinine, ejection fraction, age, creatinine phosphokinase, platelets, high blood pressure, serum sodium, anaemia, diabetes, smoking and sex. The direction of the main effects agrees with the exploratory analysis and with clinical expectation:

- High serum creatinine, which reflects impaired renal function, pushes the predicted risk upward. Low values lower it, and the largest individual contributions in the whole test set belong to this variable.
- Low ejection fraction increases risk, and a few patients with very low values receive large positive contributions.
- Higher age increases risk, with the strongest effects concentrated in the oldest patients.
- Lower serum sodium tends to increase risk, in line with the group means in Table 1.
- The presence of hypertension, anaemia, diabetes or smoking contributes a small positive amount; sex contributes almost nothing.

- CPK and platelets show non-monotone patterns, with both low and high values appearing on either side of zero, which suggests that their effect is interactive or noise-dominated at this sample size.

4.4 Local explanation

The waterfall plot for the highest-risk test patient (position 55 in the test set) illustrates how an individual score is assembled. The model's mean output over the test set is −0.42 log-odds, and the prediction for this patient is +2.93 log-odds, corresponding to a raw (uncalibrated) probability of about 0.95. The dominant contribution is an ejection fraction of 17%, which adds +2.25. Smaller positive contributions come from a platelet count of 389,000 (approximately +0.3), hypertension (+0.33), age 70 (+0.15), smoking (+0.12), serum sodium of 136 (+0.10) and diabetes (+0.08). The absence of anaemia reduces the score slightly (−0.09). Serum creatinine and sex together contribute +0.02, so for this patient the elevated risk is explained almost entirely by cardiac function rather than by renal markers.

This example shows the intended use of the method in a decision-support setting: the same patient-level breakdown can be reported alongside the risk score so that a clinician can see which measurements are responsible for it.

4.5 Model serialisation

The fitted LightGBM model and the ordered list of feature names were saved with `joblib` as `clinical_lgbm_model.pkl`. The stored estimator is the uncalibrated model, which is also the model that was explained.

5. Assessment against the project specification

Requirement	Target	Outcome	Status
Stratified 80/20 split	Required	Implemented (seed 42)	Met
Baseline vs. gradient boosting	Required	Logistic regression vs. LightGBM	Met
Hyperparameter tuning	Optuna	Fixed a priori settings only	Not done
ROC-AUC	⩾ 0.85	0.759 (0.741 calibrated)	Not met
PR-AUC	Reported	0.542 (0.511 calibrated)	Reported
Brier score	⩽ 0.15	0.186	Not met
Platt calibration	Required	Implemented	Met

Requirement	Target	Outcome	Status
Global and local SHAP	Required	Beeswarm and waterfall	Met
Serialised model	.pkl	clinical_lgbm_model.pkl	Met
FastAPI endpoint and latency	⩽ 15 ms	Not implemented, not measured	Not done
Dockerised deployment	Required	Not implemented	Not done

Table 4. Status of each deliverable and threshold.

6. Discussion and limitations

Sample size. The dataset contains 299 patients and the test set only 60, of whom roughly one in three experienced the event. At this size a single additional correct or incorrect ranking of a positive case changes ROC-AUC by several percentage points, and the sampling uncertainty of every figure in this report is large. The 0.85 ROC-AUC and 0.15 Brier targets in the specification are described in the context of a dataset of 10,000 or more records, and the performance gap observed here should be read mainly as a consequence of the data available rather than of the modelling approach. Published analyses of this dataset that report higher discrimination generally either retain follow-up time as a predictor or use cross-validation on the full sample, and neither is comparable to the leakage-free hold-out used here.

Single split. Results come from one train/test partition. A repeated stratified cross-validation, with bootstrap confidence intervals for each metric, would give a more reliable comparison between the baseline and the boosted model than the point estimates in Table 2.

Absence of tuning. The LightGBM configuration was not optimised. A systematic search (for example with Optuna and cross-validated PR-AUC as the objective) could improve both discrimination and calibration, although the potential gain on 239 training cases is limited and carries a risk of overfitting the validation folds.

Calibration. Part of the poor raw calibration is attributable to the positive-class weighting, which inflates predicted probabilities by design. Training without `scale_pos_weight`, or calibrating with isotonic regression on a larger sample, are alternatives that were not tested. Calibration curves were not plotted, so the conclusions about calibration rest on the Brier score alone.

Explainability. SHAP values describe the behaviour of the fitted model and not causal effects. The attribution to ejection fraction and serum creatinine is clinically coherent, but the instability visible in the CPK and platelet patterns indicates that secondary

attributions should not be over-interpreted. In addition, the explanations refer to the uncalibrated model, so the log-odds scale does not map exactly to the calibrated probabilities a service would return.

Target definition. The specification describes a 30-day readmission target, whereas this dataset records death during a variable follow-up period (4 to 285 days). The task is therefore mortality risk stratification, and the results cannot be transferred directly to a readmission setting.

7. Conclusion

A complete pipeline for clinical risk estimation was built on the Heart Failure Clinical Records dataset, from exploratory analysis and leakage control through model comparison, probability calibration, SHAP-based explanation and model serialisation. Gradient boosting offered a small advantage in ROC-AUC over the logistic regression baseline (0.759 against 0.734) without a clear advantage in PR-AUC, and Platt scaling gave a modest improvement in the Brier score (0.192 to 0.186). The explanations are clinically plausible, with renal function, left-ventricular ejection fraction and age as the principal determinants of predicted risk. The specification's numerical targets were not attained on this dataset, and the deployment phase remains to be completed. The next steps are repeated cross-validation with confidence intervals, a tuned and recalibrated model, calibration plots, and implementation of the FastAPI service with measured latency.

References

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